

Problem 10

Evaluate the integral:

$$I = \int \frac{x^2 + 2}{x^2 + 1} dx.$$

Solution

To solve the given integral, we decompose the fraction and evaluate it step by step.

Step 1: Split the fraction

Rewrite the integrand as:

$$\frac{x^2 + 2}{x^2 + 1} = \frac{x^2 + 1}{x^2 + 1} + \frac{1}{x^2 + 1}.$$

Simplify the terms:

$$\frac{x^2 + 2}{x^2 + 1} = 1 + \frac{1}{x^2 + 1}.$$

Thus, the integral becomes:

$$I = \int \left(1 + \frac{1}{x^2 + 1} \right) dx.$$

Step 2: Split the integral

Split the integral into two simpler integrals:

$$I = \int 1 dx + \int \frac{1}{x^2 + 1} dx.$$

Step 3: Evaluate each integral

1. The first integral is straightforward:

$$\int 1 dx = x.$$

2. The second integral, $\int \frac{1}{x^2 + 1} dx$, is a standard integral:

$$\int \frac{1}{x^2 + 1} dx = \arctan(x).$$

Step 4: Combine the results

Combine the results of the two integrals:

$$I = x + \arctan(x) + C,$$

where C is the constant of integration.

Final Answer

The solution to the integral is:

$$I = x + \arctan(x) + C.$$